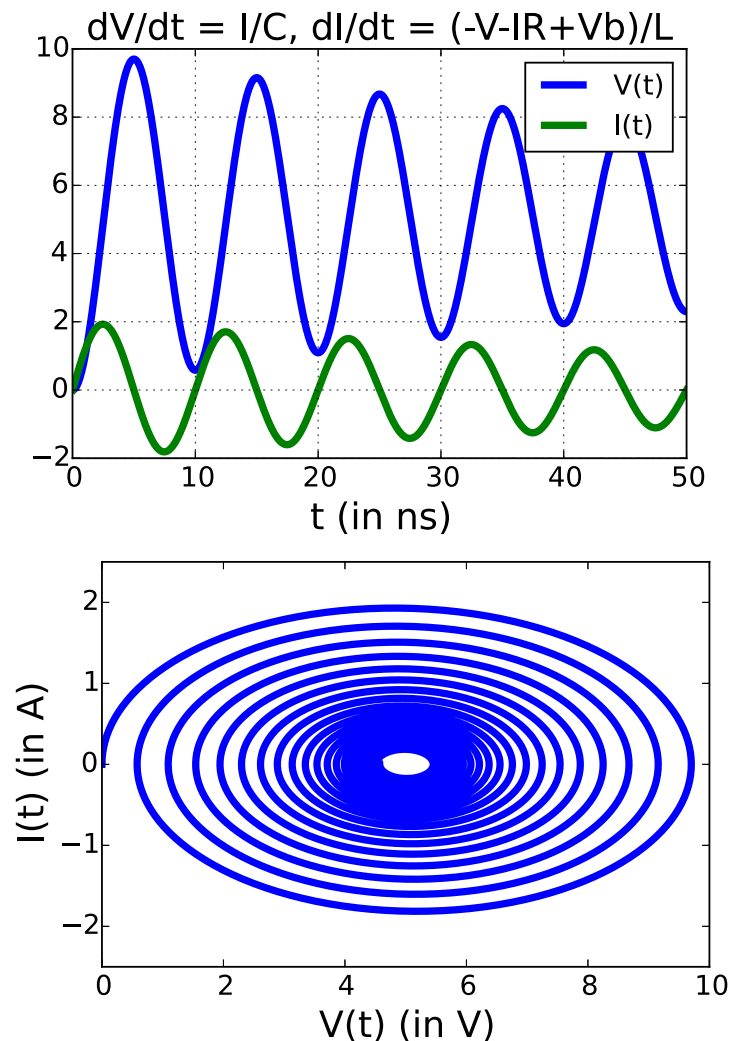




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Validation

Bookmark this page



On the last page, you have calculated for what values this LC-oscillator will oscillate with a frequency of **100 MHz**:

$V_B = 5\text{ V}$, $L = 4\text{ nH}$, $R = 0.1\ \Omega$, and $C = 0.633\text{ nF}$.

In the next exercises you change the resistance R and calculate what happens analytically.

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1/1 point (graded)

Keep $L = 4$ nH and $C = 0.633$ nF, but increase the resistance to , $R = 10 \Omega$.

For $R = 10$ the eigenvalues of the Jacobian matrix $J \left(\begin{bmatrix} V_{eq} \\ I_{eq} \end{bmatrix} \right)$ are

☒ $\lambda \approx (-1.25 \pm 1.081) \times 10^9$ ✓

☐ $\lambda \approx (-1.25 \pm 2.161) \times 10^9$

☐ $\lambda \approx (-1.25 \pm 1.081i) \times 10^9$

☐ $\lambda \approx (-1.25 \pm 2.161i) \times 10^9$

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

For $L = 4$ nH, $C = 0.633$ nF, and $R = 10 \Omega$.

For $R = 10$ the equilibrium (V_{eq}, I_{eq}) is of the type

☒ Stable node ✓

☐ Unstable node

☐ Saddle point

☐ Stable spiral point

☐ Unstable spiral point

☐ Circle point

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Exercise C

1/1 point (graded)

(For $L = 4 \text{ nH}$, $C = 0.633 \text{ nF}$) The type of the equilibrium ($V_{\text{eq}}, I_{\text{eq}}$) changes somewhere between $R = 0.1 \Omega$ and $R = 10 \Omega$. Determine analytically for what value of R the type changes.

The "tipping point" for R lies at
Give your answer in 2 decimals.

✓ Answer: 5.03

Submit

You have used 1 of 1 attempt

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Exercise D

1/1 point (graded)

Now we will focus on one special case, the one of an ideal circuit with $R = 0$.

The eigenvalues for $R = 0$ are:

☐ Real and negative.

☐ Real and zero.

☐ Real and positive.

☐ Complex with negative real parts.

☒ Complex with zero real parts. ✓

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You have used 1 of 1 attempt

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Exercise E

1/1 point (graded)

In earlier exercises, you have calculated the eigenvalues for $R = 0.1 \Omega$ and $R = 10 \Omega$ and $R = 0.0 \Omega$. In the Validation of §3.2, you have simulated the solutions for the same values with Euler's method. Do the eigenvalues and the simulated results agree?

☐ Yes.

☒ No, specifically for $R =$ they do not. Answer: 0 

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

Apparently for one value of R something is amiss... What could be the cause?

The linearisation gives eigenvalues that indicate a circle point, so in the phase plane we should see an ellipse around the equilibrium $(5, 0)$. However, when you plot the solutions you obtained with Euler's method, you see an unstable spiral point.

The simulated results from are in this case wrong. Euler's method does not work for systems with purely imaginary parts (or for eigenvalues with positive real parts). An explanation why it does not work, is outside the scope of this course, but it should be clear that we should use another time integration technique for purely imaginary eigenvalues.

One of these techniques is the fourth-order Runge-Kutta Method, which has the following equations for a differential equation $\frac{d\vec{X}}{dt} = \vec{F}(t, \vec{X})$:

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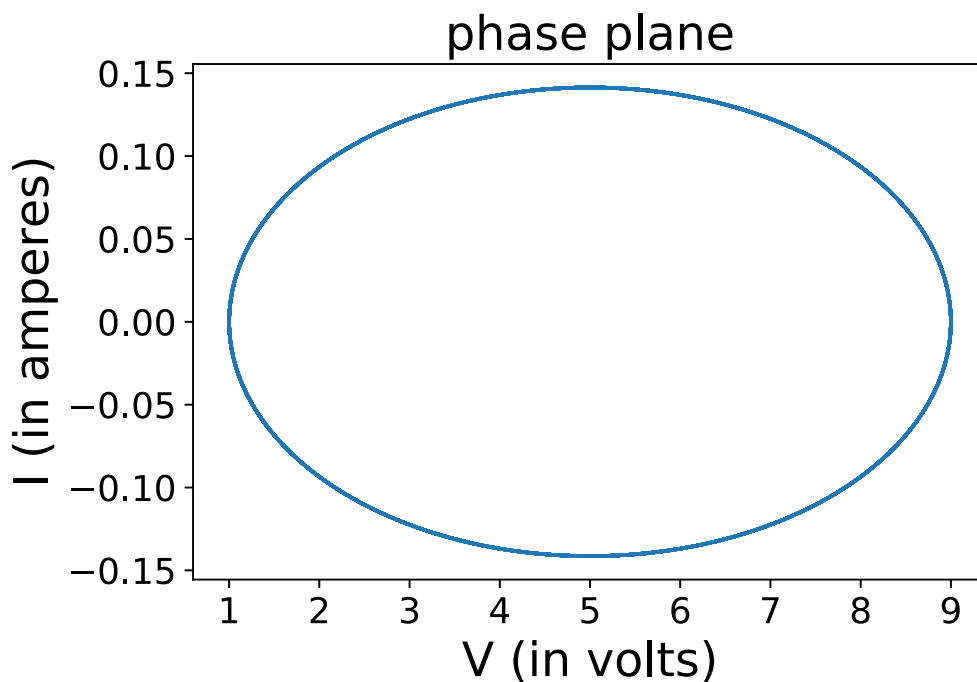
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$$\begin{aligned}
 \vec{k}_1 &= \vec{F}(n\Delta t, \vec{X}(n\Delta t)), \\
 \vec{k}_2 &= \vec{F}\left((n + \tfrac{1}{2})\Delta t, \vec{X}(n\Delta t) + \tfrac{\Delta t}{2}\vec{k}_1\right), \\
 \vec{k}_3 &= \vec{F}\left((n + \tfrac{1}{2})\Delta t, \vec{X}(n\Delta t) + \tfrac{\Delta t}{2}\vec{k}_2\right), \\
 \vec{k}_4 &= \vec{F}((n + 1)\Delta t, \vec{X}(n\Delta t) + \Delta t\vec{k}_3), \\
 \vec{X}((n + 1)\Delta t) &= \vec{X}(n\Delta t) + \tfrac{\Delta t}{6}(\vec{k}_1 + 2\vec{k}_2 + 2\vec{k}_3 + \vec{k}_4).
 \end{aligned}$$

As you can see, this method takes a lot more work to program and calculate, so we will not ask you to use this method. But you are of course allowed to. ☺

Below you can see that this more advanced method works well for the case $R = 0$.



Don't give up on Euler's method just yet however - for many systems it a great tool. Just be aware that for some specific problems, you could need other specific tools.

On the next page, a place for discussion.

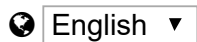
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